



Fig. 3: (a) Schematic diagram of the demonstration thermoacoustic refrigerator and (b) cross-section of the stack showing how the film layers were separated by fishing line (Courtesy: AAPT)

tions as well as refrigerators.

Also, in the future, there might be miniature cooling devices directly on computer chips. Researchers know they need a material with low thermal conductivity and high electrical conductivity, which has led them to look at compounds of heavy elements like lead, antimony, bismuth and tellurium. The search is focused on uniform bulk materials, which can be prepared in large amounts by traditional synthetic methods, and on compositionally modulated films, which require expensive nanofabrication.

## Thermoacoustic refrigeration: Sound cooling

Thermoacoustic hot-air engines (Sonic heat pump and refrigeration, or thermoacoustic heat pump and refrigeration), of which nearly all are thermoacoustic stirling engines, use high-amplitude sound waves in a pressurised gas to pump heat from one place to another, or use a heat temperature difference to induce sound, which can be converted into electricity with high efficiency, with a (piezoelectric) loudspeaker. The most efficient thermoacoustic devices built to date have a relative Carnot effi-

ciency approaching 40 per cent, which is comparable with low-end domestic vapour compression systems (high-end compressors have efficiencies up to 65 per cent) and heat engines are in most cases superior to automotive internal combustion engines.

Thermoacoustic refrigerators, now under development, use sound waves strong enough to make your hair catch

fire. But this noise is safely contained in a pressurised tube. If the tube gets shattered, the noise would instantly dissipate to harmless levels. As it conducts heat, such intense acoustic power is a clean, dependable replacement for cooling systems that use ozone-destroying CFCs.

Skeptics say current thermoacoustic designs are inefficient compared to conventional refrigeration systems, but scientists continue to improve their invention, which requires only one moving part in the form of a loudspeaker, which is more dependable than CFC-type refrigerators. Someday, household refrigerators and air-conditioners might be powered by loudspeakers blasting sound thousands of times more intense than the sound in concerts.

A thermoacoustic refrigerator works with customised loudspeakers attached to cylindrical chambers filled with inert, pressurised gases such as xenon and helium. At the opposite end of the tubes are tightly wound 'jelly rolls' made of plastic film glued to an ordinary fishing line. When the loudspeakers blast at 180 decibels, an acoustic wave resonates in the chambers. As gas molecules begin dancing frantically in response to the sound, they are compressed and heated,

with temperatures reaching a peak at the thickest point of the acoustic wave. That's where the super-hot gas molecules crash into the plastic rolls. After transferring their heat to the stack, the sound wave causes the molecules to expand and cool. Each one of these oscillating molecules acts as a member of a 'bucket brigade', carrying heat towards the source of the sound.

Inside the so-called 'bucket brigade' a gas parcel is compressed and heated by the sound wave that deposits some of its heat to the stack. The sound wave then expands and cools the gas parcel so that the gas can absorb heat from the stack and cool it. The parcels absorb heat from the cold exchanger and pass it along the stack.

## Applications abound

So we can imagine a future with compressor-free, solidstate refrigerators. Cold temperatures offered by solid-state refrigeration technologies can be tapped for chilling refrigerators, bedrooms, cars, or electronic components on satellites and inside computers. Besides air-conditioning and refrigeration units, applications could include heating or cooling of a variety of clothing including cooling of protective gear for fire fighters, heating of mittens and socks or shoes for athletes, sportsmen and law enforcement officers, and even cooling of mascot and cartoon character costumes. Another application would be in electronics, where small amounts of polymers could effectively cool overheating circuit boards and allow closer packing and therefore smaller devices.

Initially, these could be implemented in various types of niche applications—scale refrigerating plants, soda machines or places where a noise-free environment is important—before ultimately spreading to the rest of the society. ●

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